Sequences of real numbers

Remarkable sequences:

- $c_n = \left(1 + \frac{1}{n}\right)^n \nearrow e = 2{,}7182$
- $E_n = 1 + \frac{1}{1!} + \frac{1}{2!} + \ldots + \frac{1}{n!} \nearrow e$
- $\gamma_n = 1 + \frac{1}{2} + \ldots + \frac{1}{n} - \ln n \searrow \gamma \cong 0{,}57$ (Euler's ct)

$$\lim_{n\to\infty}\left(1 + \frac{1}{2} + \ldots + \frac{1}{n}\right)$$

$H_n = 1 + \frac{1}{2} + \ldots + \frac{1}{n}$ (Harmonic sequence)

$$\lim_{n\to\infty} H_n = \lim_{n\to\infty}\left(\gamma_n + \ln n\right) = \gamma + \infty = +\infty$$

Pol Calculate:

a) $$\lim_{n\to\infty}\left(\frac{1}{n+1} + \frac{1}{n+2} + \ldots \frac{1}{2n}\right)$$

$$\lim_{n\to\infty}\Big(\underbrace{1 + \frac{1}{2} + \ldots + \frac{1}{n} + \frac{1}{n+1} + \ldots + \frac{1}{2n} - \ln 2n}_{\gamma_{2n}}\Big)$$

$$-\underbrace{\left(1 + \frac{1}{2} + \ldots + \frac{1}{n} - \ln n\right)}_{\gamma_n} + \ln 2n - \ln n$$

$$\lim_{n\to\infty}\left(\gamma_{2n}-\gamma_n+\ln\left(\frac{2n}{n}\right)\right)=\gamma-\gamma+\ln 2=\ln 2$$

b) $$\lim_{n\to\infty}\left(e^{1+\frac{1}{2}+..+\frac{1}{n+1}}-e^{1+\frac{1}{2}+...+\frac{1}{n}}\right)$$

$$\lim_{n\to\infty} e^{1+\frac{1}{2}+..+\frac{1}{n}}\left(e^{\frac{1}{n+1}}-1\right)\cdot\frac{1}{\frac{1}{n+1}}$$

$$\lim_{n\to\infty}\left(e^{\gamma_n+\ln n}\cdot\frac{1}{n+1}\right)$$

$$\lim_{n\to\infty}\left(e^{\gamma_n}\cdot\frac{e^{\ln n}}{n+1}\right)=\lim_{n\to\infty} e^{\gamma_n}\cdot\frac{n}{n+1}=e$$

Important criterias for limits of sequences

I Stolz-Cesaro Th. $\left(\frac{\infty}{\infty}\right)$

Let $(a_n)_{n\geq 1}$, $(b_n)_{n\geq 1}$ be 2 sequences of real numbers with the following properties:

1) $(b_n)_{n\geq 1}$ is strictly monotonic and unbounded

2) $\exists \lim_{n\to\infty}\frac{a_{n+1}-a_n}{b_{n+1}-b_n}=\ell\in\overline{\mathbb{R}}$

Then $\lim_{n\to\infty}\frac{a_n}{b_n}=\ell$

Pr 2 Calculate:

a) $\lim_{n\to\infty}\frac{1+\frac{1}{2}+\ldots+\frac{1}{n}}{\ln n}$, $n\geq 2$

1) $\ln n \nearrow$

2) $$\lim_{n\to\infty} \frac{1+\frac{1}{2}+\ldots+\frac{1}{n}+\frac{1}{n+1}-1-\frac{1}{2}-\ldots-\frac{1}{n}}{\ln(n+1)-\ln n}$$

$$\lim_{n\to\infty} \frac{1}{(n+1)\cdot\ln\left(\frac{n+1}{n}\right)} = \lim_{n\to\infty} \frac{1}{\ln\left(\frac{n+1}{n}\right)^{n+1}} = \frac{1}{\ln e}$$

$= 1$

$$\lim_{n\to\infty} \frac{1+\frac{1}{2}+\ldots+\frac{1}{n}}{\ln n} = 1$$